Sardor Sobirov
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https://sardors-portfolio.vercel.app

Education

University of California San Diego (UCSD) La Jolla, CA
Bachelor of Science in Data Science *Jun 2026*

- **Relevant Coursework**: Machine Learning Algorithms, Deep Learning and Neural Networks, Probabilistic AI and Graphical Models, Data Mining and Statistical Learning, Distributed Data Systems and Scalable Analytics.

Skills

Languages and ML: Python, SQL, JavaScript/TypeScript, pandas, NumPy, scikit-learn, PyTorch, LightGBM, Hugging Face Transformers.
Data Engineering: PostgreSQL, BigQuery, API ingestion, ETL/ELT, JSON/JSONL, Parquet, NetCDF, data validation, reproducible pipelines.
Analytics and Visualization: A/B testing, hypothesis testing, KPI design, Power BI, Tableau, React/Next.js, Streamlit, D3.js, MapLibre.

Experience

Geohazard Data Science Intern | U.S. Department of the Interior (National Park Service)
Mount Rainier National Park, WA *Sep 2026 – Jun 2027*

- Develop geospatial digital twins of Mount Rainier river and glacier systems from LiDAR, satellite imagery, and field sensors using Python and ArcGIS Pro to monitor geomorphic change.
- Build statistical and machine-learning workflows for seismic, RFID/BLE tracer, and streamflow data to quantify sediment transport and detect geohazard signals for the NPS Imminent Threats Program.

Data Science Research and Development Intern | Cotality
San Diego, CA *Sep 2025 – Mar 2026*

- Built a React/TypeScript data-quality dashboard with MapLibre GL and D3.js, turning notebook-generated CSV/JSON artifacts into reviewer-facing views for Cotality's wildfire-risk anomaly investigation.
- Processed 2.4M+ property records across 58 California counties with BigQuery GIS, SQL, H3 indexing, spatial joins, buffers, and centroid geometry checks to produce QA/QC tables for model review.

Data Engineer Intern | Apptics AI
Walnut Creek, CA *Jun 2025 – Sep 2025*

- Built a reproducible Python ETL pipeline ingesting daily paid media performance data from the Meta Ads API into PostgreSQL for 5 DTC brands, consolidating $1K-$10K/day per brand spend into a single source of truth for reporting.
- Designed campaign, ad set, and ad-level schemas; standardized 25+ SQL KPIs and validation checks so report issues could be traced from source API pulls through Power BI outputs.

Research and Development Intern | United States Department of Defense
La Jolla, CA *Sep 2024 – Dec 2024*

- Conducted 50+ stakeholder interviews using Lean LaunchPad methodology to translate ambiguous user needs into a prioritized product roadmap, ensuring research and development investments were tied to concrete operational use cases.
- Designed a portfolio analytics dashboard in Excel and Power BI mapping technology maturity signals to DoD use cases; presented findings to cross-functional leadership to inform multi-million dollar transition decisions.

Projects

Crop Belt Modeling | Dashboard | Paper | GitHub

- Built an end-to-end ML/data pipeline for the 13-state Corn Belt, integrating USDA CDL, MODIS NDVI, and NASA SMAP data into NetCDF cubes, Parquet tables, metadata sidecars, and dashboard-ready artifacts.
- Modeled crop phenology, rotation, soil-moisture anomalies, and crop prediction using HSGP, Dirichlet-Multinomial, NIG Bayesian scoring, and LightGBM; achieved 79.2% accuracy with SHAP and ablation diagnostics.

Wildfire Property Intelligence | Dashboard | Paper | GitHub

- Created a statistical anomaly-review workflow using Empirical Bayes, neighbor divergence, classifier two-sample tests, Moran's I, and static JSON exports for a React/TypeScript dashboard.
- Converted sparse-data artifacts, land-cover differences, and inconsistent category labels into reviewable data-quality signals for wildfire property-risk modeling.

Activities

- **Cornerstone Community Consultants, Developer Lead**: converted small-business client briefs and designer mockups into implementation tickets, acceptance criteria, and shipped website changes; coordinated developers through review cycles.
- **Computer Science and Engineering Society, Technical Project Lead**: modeled member, event, and outreach data flows for club web systems, including login/access logic, SQL-style schemas, and Mailchimp audience coordination.